ERNEST NG

ernest@cs.cornell.edu | ngernest.github.io | github.com/ngernest

EDUCATION

Cornell University
PhD in Computer Science — 2024 - Present

University of Pennsylvania
MSE in Computer Science — 2022 - 2023
BA in Computer Science & Mathematics (*summa cum laude*) — 2018 - 2022

RESEARCH EXPERIENCE

Cornell University — August 2024 - Present
Graduate Research Assistant

- Developing a language for specifying hardware communication protocols and inferring high-level transactions from signal traces (under submission)
- Advised by Prof. Adrian Sampson and Dr. Kevin Laeufer

University of Pennsylvania — December 2022 - August 2024
Graduate Researcher

- Developed Mica, an OCaml compiler extension that automates differential testing of OCaml modules
  - Presented paper at the OCaml Users and Developers Workshop, co-located with ICFP 2024
  - Received Second Prize at the ICFP 2023 Student Research Competition (poster)
  - Advised by Profs. Benjamin Pierce and Harrison Goldstein

INDUSTRY EXPERIENCE

Amazon Web Services — May - November 2026
Applied Scientist Intern — *Boston, MA*

- Developed infrastructure in the Lean proof assistant for proving property-based testing generators correct, with applications to the Strata intermediate language (submission in preparation)
- Advised by Prof. Mike Hicks and Dr. Joshua Cohen

Amazon Web Services — May - August 2025
Applied Scientist Intern — *Boston, MA*

- Developed Specimen, a metaprogramming tool which derives correct-by-construction random generators based on inductive relations specified in Lean, advised by Prof. Mike Hicks and Dr. Cody Roux

TEACHING EXPERIENCE

Cornell University
Teaching Assistant

- CS 4610: Formal Verification (Rocq) — Spring 2026
- CS 3410: Computer Systems (C, RISC-V) — Fall 2025

University of Pennsylvania
Teaching Assistant

- CIS 3800: Operating Systems (C) — Fall 2023
- *Software Foundations* Rocq course (Penn Programming Languages REU) — Summer 2023
- CIS 1940: Introduction to Haskell — Spring 2023
- CIS 5450: Big Data Analytics (Head Teaching Assistant) — Spring - Fall 2022
- CIS 2400: Computer Systems (C) — Fall 2022
- CIS 5500: Databases — Summer 2022

ACADEMIC ACTIVITIES AND SERVICE

- Finalist, Qualcomm Innovation Fellowship — 2026
- SIGPLAN AV Student Volunteer (ICFP 2023 / 2024, POPL 2024) — 2023 - 2024
- Cornell, Maryland, Max Planck Pre-Doctoral Research School (CMMRS) Travel Award — August 2023
- Oregon Programming Languages Summer School (OPLSS) Fellowship Grant — June 2023